

Train a Regression Model

Project: Responsible Plant Classifier

1. Train a Regression Model

Teach the learning cycle

Primary target: Fit a line from a declared training candidate set

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate MAE MSE and RMSE

Reference example, not the only acceptable wording: Teacher explanation: $(1+2+3)/3=2$ cm.

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish numeric prediction from classification

Reference example, not the only acceptable wording: Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The training-majority rule always predicts clear. Matrix $\begin{bmatrix} 16 & 0 \\ 4 & 0 \end{bmatrix}$ gives $16/20=80$ percent, $FP=0, FN=4$. It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training points $(x,y)=(1,3),(2,5),(3,7)$. Compare $y=x+1$ and $y=2x+1$ by training MAE; select one, then predict held-out $x=4$, actual=10. Explain what was fitted.

Concept to explain

A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

Term · Meaning

Term: coefficient

Meaning: learned relationship weight

Term: intercept

Meaning: prediction when inputs are zero

Term: slope

Meaning: predicted change per input unit

Term: generalization

Meaning: performance on unseen cases

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on fitting a line. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Square each residual, sum, then divide by the number of evaluated examples.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain train a regression model using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Run a starter regression and explain slope, prediction, residual, and MAE.

MODEL IDEA A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Model predicts height = $4 + 2 \times \text{weeks}$. Slope 2 means two predicted centimeters per additional week in this data range.

Draw a fresh data $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Model predicts height = $4 + 2 \times \text{weeks}$.

STEP 2 Slope 2 means two predicted centimeters per additional week in this data range.

STEP 3 At week 5, prediction is 14 cm; if actual is 16, residual is +2 cm.

STEP 4 Evaluate unseen plants and compare test MAE with the median baseline.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Supplied fitted rule: height = 2 times age + 1. Held-out observations are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Exact explanation to model

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Explain the added worked example

A line predicts y as $m \times x + b$. In this classroom search we compare a small declared set of slopes and intercepts; this is not a full least-squares solver over all possible lines. Use training error to choose a candidate, then freeze it before validation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines fitting a line.

Training pairs are (0, 1),(1, 3),(2, 5). Candidate A has $m = 1$ $b = 1$, B $m = 2$ $b = 0$, C $m = 2$ $b = 1$. Predictions are A [1, 2, 3], B [0, 2, 4], C [1, 3, 5]. Training squared-error sums are 5, 3, 0 respectively, so choose C. Ties would use the earlier listed candidate.

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Key terms

Term · Meaning

Term: Slope

Meaning: Change in predicted target per unit change in x

Term: Intercept

Meaning: Predicted target when x is zero

Term: Candidate set

Meaning: The declared model choices searched

Term: Fitting

Meaning: Selecting model values using training evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Fitting a line at a glance

Part · Evidence or action · Interpretation

Part: Line A

Evidence or action: Slope 1 intercept 1

Interpretation: Training SSE 5

Part: Line B

Evidence or action: Slope 2 intercept 0

Interpretation: Training SSE 3

Part: Line C

Evidence or action: Slope 2 intercept 1

Interpretation: Training SSE 0

For separate validation pairs (3, 6), (4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Predict at x 3 and x 4 and calculate MAE/MSE. Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train a Regression Model

Fit a toy growth line from training pairs, then leave evaluation labels out of fitting.

The facts we will use

Training(day, height):(0, 1),(1, 3),(2, 5). Candidates $y=2x+1$ and $y=x+1$. Evaluation pair:(3, 6).

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/22

Train a Regression Model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Calculate training fit

$2x+1$ predicts 1, 3, 5 with SSE0; $x+1$ predicts 1, 2, 3 with SSE5.



2. Select line

Choose $2x+1$ under minimum training SSE.



3. Evaluate once

At day3 predict 7 versus observed 6: absolute error 1 cm.

CHANGE ONE THING

Evaluation observation becomes 8 instead of 6.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Calculate training fit

What happens and why: $2x+1$ predicts 1, 3, 5 with SSE0; $x+1$ predicts 1, 2, 3 with SSE5.

Step: 2. Select line

What happens and why: Choose $2x+1$ under minimum training SSE.

Step: 3. Evaluate once

What happens and why: At day3 predict7 versus observed6: absolute error1 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

Change one thing in the pictured case

Change: Evaluation observation becomes8 instead of6.

Prediction: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Reason: Prediction7 remains frozen; changed observation8 reverses the error direction.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Supplied fitted rule: $\text{height}=2 \text{ times age}+1$. Held-out observations are (3,8),(4,9).

Write the question to explain: Predict and evaluate; distinguish inference.

Reveal after predictions: Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Keep for an independent attempt: A regression model has validation MAE 2.4 cm; a training-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Predictions 7, 9; residuals -1, 1; MAE 1 MSE 1. Exit response: Training covers a limited range; the relationship and conditions may change outside it.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: How is MSE calculated? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify slope in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: trace candidates; score validation. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between slope and intercept. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Fit the training constant and score it on validation.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Fit a line from a declared training candidate set; Separate training loss from validation comparison. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Predict at $x = 3$ and $x = 4$ and calculate MAE/MSE.

Teacher check: Teacher explanation: Predictions 7, 9; residuals -1, 1; MAE 1 MSE 1.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Fitting a line. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: A 5 B 3 C 0; choose m 2 b 1.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step

for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

No under this fitting plan. Keep evaluation evidence separate and report the frozen line's 1 cm error. Changed-case check: Absolute evaluation error stays 1 cm, but residual changes from -1 to +1. Prediction 7 remains frozen; changed observation 8 reverses the error direction.

Changed-case reasoning: Absolute evaluation error stays 1 cm, but residual changes from -1 to +1. Prediction 7 remains frozen; changed observation 8 reverses the error direction.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Regression Run: trace code, interpret coefficients with units, predict unseen values, and compare MAE with baseline.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A positive slope proves one variable causes the other.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: A 5 B 3 C 0; choose m 2 b 1.

A stronger claim makes a stronger project: Training covers a limited range; the relationship and conditions may change outside it.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: SSE 5, 3, 0 selects m 3 b 2. Predictions 11, 14; residuals -1, 1; MAE 1.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Independent question

A regression model has validation MAE 2.4 cm; a training-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

Expected independent answer

The baseline is better because lower MAE is preferable: $2.0 < 2.4$ cm. Investigate the model before claiming improvement.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A trained model necessarily outperforms a simple baseline.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

First model errors are 1,2,3, MAE=2; second errors are 0,0,0, MAE=0. Select $2x+1$. Test prediction=9 and absolute error=1. This selects among two supplied candidates; it is not a derivation of unrestricted linear-

regression fitting.

Explanations for the additional workbook examples

W1 Trace candidates

Give the three training SSE values and selected line.

Teacher explanation: A 5 B 3 C 0; choose m 2 b 1.

W2 Score validation

Predict at x 3 and x 4 and calculate MAE/MSE.

Teacher explanation: Predictions 7, 9; residuals -1, 1; MAE 1 MSE 1.

W3 Compare the baseline

Fit the training constant and score it on validation.

Teacher explanation: Training mean 3; residuals 3, 7; MAE 5 MSE 29.

W4 Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

Teacher explanation: SSE 5, 3, 0 selects m 3 b 2. Predictions 11, 14; residuals -1, 1; MAE 1.

W5 Limit extrapolation

Why is an exact training line not proof of correct predictions at x 100?

Teacher explanation: Training covers a limited range; the relationship and conditions may change outside it.

Assess independent transfer

Selected case: Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

Selected case answer

Teacher explanation: SSE 5, 3, 0 selects m 3 b 2. Predictions 11, 14; residuals -1, 1; MAE 1.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Add the regression experiment to the project appendix only as a separate, clearly limited model.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 23: Hold Out Test Data. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use slope and intercept in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: SSE 5, 3, 0 selects m 3 b 2. Predictions 11, 14; residuals -1, 1; MAE 1. Fresh retry training (0, 0), (1, 2), (2, 4), candidates m 1 b 0 and m 2 b 0: SSE 5 and 0. At validation (3, 7), chosen prediction 6 gives residual 1. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Validation and model selection. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/23, ai-4/25, ai-4/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Fit and test a straight-line regression model

Fit and test a straight-line regression model

Estimate slope and intercept from training pairs using least squares. Ages are weeks and heights are cm. Test targets are excluded from fitting; training ages must include distinct values.

```
"""Fit a line to training pairs and evaluate a separate example."""

train = [(1, 3), (2, 5), (3, 7)]

if not train:
    raise ValueError("Training examples are required")

n = len(train)

x_mean = sum(x for x, y in train) / n
y_mean = sum(y for x, y in train) / n
num = sum((x - x_mean) * (y - y_mean) for x, y in train)
den = sum((x - x_mean) ** 2 for x, y in train)

if den == 0:
    raise ValueError("Distinct training ages are required")

slope = num / den
intercept = y_mean - slope * x_mean
test = [(4, 10)]

if not test:
```

```

raise ValueError("Test examples are required for MAE")

preds = [slope * x + intercept for x, y in test]

mae = sum(abs(y - pred) for pred, (x, y) in zip(preds, test)) / len(test)

print(slope, intercept, preds, mae)

```

Expected output: 2.0 1.0 [9.0] 1.0

The fitted line is height = 2 times age + 1. At four weeks it predicts 9 cm; the observed height of 10 cm gives an absolute error and one-case MAE of 1 cm. Fitting a line and applying a supplied line are different activities. One held-out case is very limited evidence.

Independent code change

Change the held-out height from 10 to 8 without changing the training data. Predict the fit, prediction and MAE.

Teacher answer: Slope 2, intercept 1, prediction 9 cm and MAE 1 cm are unchanged. The residual sign changes under the actual-minus-predicted convention.

8. Prepare, teach and assess

Teaching target: Fit a line from a declared training candidate set

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38:

connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Predict at $x = 3$ and $x = 4$ and calculate MAE/MSE.

Reasoned practice answer: Teacher explanation: Predictions 7, 9; residuals -1, 1; MAE 1 MSE 1.

Check every learner before discussion: Use the worked case for Fitting a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: A 5 B 3 C 0; choose m 2 b 1.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Evaluation observation becomes 8 instead of 6. Require a prediction and a reason before comparing. Expected result: Absolute evaluation error stays 1 cm, but residual changes from -1 to +1. Reason: Prediction 7 remains frozen; changed observation 8 reverses the error direction.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

9. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.